CONTROL X MASTER SPEC v2.2

CONTROL_X_MASTER_SPEC.md1. Project OverviewProject VisionTo construct a luxury digital web application for ControlX, establishing an interactive benchmark for high-end digital agency platforms. The experience synthesizes editorial typography with real-time physical WebGL optics, glassmorphism UI layers, and fluid dynamics.

```
┌──────────────────────────────────────────────────────────────┐
│ DOM UI OVERLAY LAYER │
│ [Navbar] [Typography] [Forms] [Glass Cards] [CTA] │
└──────────────────────────────┬───────────────────────────────┘
│ Alpha / Depth Blur Blending
┌──────────────────────────────┴───────────────────────────────┐
│ THREE.JS / WEBGL CANVAS │
│ [3D Glass Mesh] [Physical Lights] [GPUParticleSystem] │
└──────────────────────────────┬───────────────────────────────┘
│ Framebuffer Pass
┌──────────────────────────────┴───────────────────────────────┐
│ GLSL POST-PROCESSING STACK │
│ [Chromatic Aberration] [Unreal Bloom] [Vignette Pass] │
└──────────────────────────────────────────────────────────────┘
```

MissionTranslate high-level creative vision into a technical framework. The interface operates as a physical studio stage where 3D refractive assets, dynamic camera moves, and UI elements react to scroll momentum and pointer vectors.Brand PhilosophyPrecision Aesthetics: Clean editorial structure aligned with optical physical realism.Material Fidelity: Glass elements simulate physical properties: index of refraction ($IOR = 1.52$), internal absorption, specular dispersion, and chromatic aberration.Fluid Motion: Spatial motion follows physical mass, spring mechanics, and damped inertia.Creative DirectionColor Palette: Warm neutral studio tones (#EBE9E1) contrasting with emerald glass refractions (#0F8259).Typography: Dual-font hierarchy—Classic Serif (Playfair Display) for editorial headings, paired with Sans-Serif (Plus Jakarta Sans) for UI controls and numerical data.Depth Architecture: Multi-layered visual stack using WebGL depth buffers, glass refraction passes, and dynamic backdrop filters.Target Audience & Business GoalsTarget: Executive clientele, enterprise partners, luxury brands, and creative directors.Business Goal: Establish technical expertise through design execution, convert high-value leads via interactive inquiry touchpoints, and demonstrate digital craftsmanship.User Journey Map

```
┌─────────────────┐  ┌─────────────────┐  ┌─────────────────┐
│ 01. INTRO │───> │ 02. SERVICES │───> │ 03. PORTFOLIO │
│ Emerald Assembly│ │ Glass Grid Cards│ │ Pinned Showcase │
└─────────────────┘  └─────────────────┘  └─────────────────┘
│
```

```
┌──────────────────┐  ┌──────────────────┐  │
│ 05. OUTRO │ <───│ 04. PROCESS │ <───────────────────┘
│ Contact & Footer│ │ Node Rail Track │
└──────────────────┘  └──────────────────┘
```

Success MetricsPerformance: Constant 60 FPS on desktop (DPR 2.0) and mobile (DPR 1.0).First Contentful Paint (FCP): $< 0.8\text{s}$.Time to Interactive (TTI): $< 1.8\text{s}$.Cumulative Layout Shift (CLS): 0.00.2. Design LanguageLuxury & Editorial AestheticsThe visual hierarchy uses asymmetrical layouts, large display headers, and deliberate line heights. Spatial layout relies on high negative space ratios ($> 55\%$) to maintain visual hierarchy.Glassmorphism OpticsGlass elements use backdrop blur filtering alongside WebGL-rendered refraction maps.CSS Backdrop Blur: $20\text{px} - 24\text{px}$ blur radius with saturation boost (180%).WebGL Refraction: Custom MeshPhysicalMaterial with $IOR = 1.52$ (Simulating Crown Glass), 95% transmission, and internal emerald attenuation.Motion PhilosophyAll spatial animations avoid linear easing curves. Movements utilize custom cubic-bezier curves or spring physics:Standard Structural Entrance: cubic-bezier(0.16, 1, 0.3, 1) (Power3 / Quartz Curve).Elastic UI Interactions: cubic-bezier(0.34, 1.56, 0.64, 1) (Back Out / Spring Pop).Inertia Scrolling: Damped spring lag with a smoothing lerp coefficient of 0.08.Camera & Lighting PhilosophyThe camera functions as a physical lens with a fixed focal length ($f = 50\text{mm}$, $FOV = 45^\circ$). Lighting uses a three-point studio system:Key Light: High-intensity white directional light casting directional floor shadows.Fill Light: Soft cool fill light neutralizing shadow contrast.Accent Point Light: Emerald-tinted point light placed behind glass objects to highlight refractions.3. Global Design Tokens GLOBAL TOKEN TREE

```
│
┌─────────────────────────────┼─────────────────────────────┐
│ │ │
┌───▼───┐ ┌───▼───┐ ┌───▼───┐
│COLORS │ │ TYPO │ │SPACING│
└───┬───┘ └───┬───┘ └───┬───┘
├── Primary: #EBE9E1 ├── Hero: 72px / 1.1 ├── Section: 120px
├── Text: #18191C ├── H2: 48px / 1.15 ├── Card Gap: 32px
└── Accent: #0F8259 └── Body: 15px / 1.5 └── Padding: 24px
```

Color TokensJSON{

```
"color": {
"bg": {
"primary": { "value": "#EBE9E1", "type": "color" },
"gradient_start": { "value": "#FAF9F5", "type": "color" },
"gradient_end": { "value": "#E2DFD5", "type": "color" }
},
```

"text": { "primary": { "value": "#18191C", "type": "color" }, "secondary": { "value": "#686D76", "type": "color" }, "muted": { "value": "#9E9E9E", "type": "color" } },

"accent": { "emerald": { "value": "#0F8259", "type": "color" }, "emerald_light": { "value": "#10B981", "type": "color" }, "emerald_glow": { "value": "rgba(15, 130, 89, 0.35)", "type": "color" } },

"glass": { "fill": { "value": "rgba(255, 255, 255, 0.25)", "type": "color" }, "border": { "value": "rgba(255, 255, 255, 0.40)", "type": "color" }, "shadow": { "value": "rgba(0, 0, 0, 0.06)", "type": "color" } }

}

}

Typography TokensJSON{

"typography": {

"family": { "serif": { "value": "'Playfair Display', Georgia, serif" }, "sans": { "value": "'Plus Jakarta Sans', system-ui, sans-serif" } },

"display": { "hero": { "fontSize": "72px", "lineHeight": "1.1", "fontWeight": "600", "letterSpacing": "-0.02em" }, "h1": { "fontSize": "64px", "lineHeight": "1.12", "fontWeight": "600", "letterSpacing": "-0.02em" }, "h2": { "fontSize": "48px", "lineHeight": "1.15", "fontWeight": "600", "letterSpacing": "-0.01em" } },

"ui": { "card_title": { "fontSize": "24px", "lineHeight": "1.25", "fontWeight": "600" }, "body": { "fontSize": "15px", "lineHeight": "1.5", "fontWeight": "400" }, "button": { "fontSize": "14px", "lineHeight": "1.0", "fontWeight": "700", "letterSpacing": "0.05em" }, "label": { "fontSize": "12px", "lineHeight": "1.3", "fontWeight": "600", "letterSpacing": "0.1em" } }

}

}

Spatial & Layout TokensJSON{

"spacing": { "section_padding_y": "120px", "container_max_width": "1440px", "gutter": "24px", "card_padding": "32px", "element_gap": "16px" },

"radius": { "sm": "8px", "md": "16px", "lg": "24px", "pill": "9999px" },

"z_index": { "canvas": 0, "content": 10, "navbar": 100, "cursor": 999 }

}

CSS Custom Properties Variables ExportCSS:root {

/* Colors */

--color-bg-primary: #EBE9E1;

--color-text-primary: #18191C;

--color-text-secondary: #686D76;

--color-accent-emerald: #0F8259;

--color-accent-emerald-light: #10B981;

--color-glass-fill: rgba(255, 255, 255, 0.25);

--color-glass-border: rgba(255, 255, 255, 0.40);

/* Fonts */

--font-serif: 'Playfair Display', Georgia, serif;

--font-sans: 'Plus Jakarta Sans', system-ui, sans-serif;

/* Spatial */

--container-max-width: 1440px;

--radius-md: 16px;

--radius-lg: 24px;

--radius-pill: 9999px;

/* Motion */

--ease-quartz: cubic-bezier(0.16, 1, 0.3, 1);

--ease-spring: cubic-bezier(0.34, 1.56, 0.64, 1);

}

4. Asset Manifest3D Models & WebGL GeometriesAsset IDPath / TypeVertex CountTarget TrisMaterial BindingsOptimization Strategygeo_x_shards/models/x_shards.glb24,50012,000mat_emerald_glassDraco Compression, Instanced Arraysgeo_node_discsTHREE.CylinderGeometry384192mat_node_glassGenerated via WebGL InstancedMeshgeo_tube_railTHREE.TubeGeometry2,0481,024mat_tube_glassCatmullRomCurve3 interpolationTextures & MapsTexture IDResolutionFormatColor SpacePurposetex_noise_alpha512×512PNGLinearScreen-space grain overlaytex_hdr_studio2048×1024HDRLinearEnvironment map for glass reflectionstex_normal_fluid1024×1024PNGNormal MapNormal distortion map for background liquid shader5. Object Registry OBJECT REGISTRY SHADOW MATRIX

UUID	NAME	TYPE	SCENE LAYER
obj_webgl_canvas	Canvas Root	HTMLCanvasElement	Base WebGL (z-idx:0)
obj_3d_x_assembly	Glass X Logo Mesh	THREE.Group	WebGL Scene
obj_3d_node_rail	Process Tube Rail	THREE.Mesh	WebGL Scene
obj_ui_navbar	Main Navbar	HTMLElement	DOM Layer (z-idx:100)
obj_ui_hero_title	Hero Header Text	HTMLElement	DOM Layer (z-idx:10)

obj_ui_services_grid	Services 3x2 Grid	HTMLElement	DOM Layer (z-idx:10)
obj_ui_contact_card	Glass Contact Form	HTMLElement	DOM Layer (z-idx:10)

Comprehensive Technical Declarationsobj_3d_x_assemblyUUID: c102a488-842e-4bdf-9921-992641040801Parent: scene_rootTransform Initial: Position $(0, 0, 0)$, Scale $(1, 1, 1)$, Rotation $(0, 0, 0)$.Material: mat_emerald_glass ($IOR = 1.52$, Transmission $= 0.95$).Animation State: Implosion assembly on load ($t_0 - t_1.2s$), constant idle sine wave rotation ($y = \sin(t \cdot 0.5) \cdot 0.15$).Visibility Rules: Visible during Intro ($0s - 3s$) and Footer Outro ($8s - 10s$). Hidden during middle scroll sections via shader opacity uniform.obj_ui_services_gridUUID: a84920df-7711-41e2-b883-200921811702Parent: dom_bodyLayout Structure: CSS Grid (3 Columns $\times$ 2 Rows), Gap: 24px.Glass Styling: backdrop-filter: blur(20px), background: rgba(255,255,255,0.25).Interaction State: Mouse hover induces card Z-translation ($+12\text{px}$) and border brightness shift.6. Scene Graphroot_application

```
│
├── dom_layer (z-index: 10 - 100)
│ ├── comp_navbar (fixed, top: 0)
│ │ ├── elem_brand_logo
│ │ ├── elem_nav_links
│ │ └── elem_signup_btn
│ │
│ ├── sec_hero (100vh)
│ │ ├── elem_hero_title
│ │ └── elem_hero_buttons
│ │
│ ├── sec_services (200vh)
│ │ └── comp_services_grid (3x2 Cards)
│ │
│ ├── sec_portfolio (250vh)
│ │ └── comp_horizontal_slider
│ │
│ ├── sec_process (250vh)
│ │ └── comp_node_labels_overlay
│ │
│ ├── sec_contact (120vh)
│ │ └── comp_glass_form_card
```

```
│ │
│ └── comp_cursor (fixed, z-index: 999)
│
└── webgl_layer (z-index: 0)
└── three_scene
├── sys_camera (PerspectiveCamera)
├── sys_lighting
│ ├── light_ambient
│ ├── light_directional
│ └── light_emerald_point
│
├── mesh_background_plane (ShaderMaterial)
├── mesh_x_glass_group (InstancedMesh)
├── mesh_process_rail (TubeGeometry)
└── sys_particle_emitter (GPUParticleSystem)
```

7. Camera System CAMERA MOTION CHOREOGRAPHY TRACK

```
Z-Pos
10.5 ──┐
│ ┌─────────────────────┐
8.0 ──┴────────────────────┐ │ │
└─────────────────────────┘ └────────────────────

0.0
────┼───────────────────┼─────────────────────────┼─────────────────────┼──────────────────>
Time
0s 1.5s 5.5s 8.0s 10s
```

Camera Blueprint ConfigurationType: PerspectiveCameraFocal Length / FOV: 45° FOVNear Plane: 0.1Far Plane: 1000.0Choreography KeyframesJSON[

{ "time": 0.0, "position": [0.0, 0.0, 8.0], "rotation": [0.0, 0.0, 0.0], "ease": "linear" },

{ "time": 1.5, "position": [0.0, 0.0, 8.0], "rotation": [0.0, 0.0, 0.0], "ease": "cubic-bezier(0.16, 1, 0.3, 1)" },

{ "time": 5.5, "position": [0.0, -2.5, 10.5], "rotation": [-0.05, 0.0, 0.0], "ease": "cubic-bezier(0.16, 1, 0.3, 1)" },

{ "time": 8.0, "position": [0.0, -5.0, 9.0], "rotation": [0.0, 0.0, 0.0], "ease": "cubic-bezier(0.16, 1, 0.3, 1)" },

{ "time": 10.0, "position": [0.0, -7.2, 7.5], "rotation": [0.0, 0.0, 0.0], "ease": "cubic-bezier(0.25, 1, 0.5, 1)" }

]

8. Lighting System LIGHTING VECTOR ORIENTATION

[Key Light]

(5.0, 10.0, 7.0)

│ \

│ \ Direct Rays

│ ▼

[Fill Light] ──────────> [Target: Glass Mesh] <────────── [Point Light]

(-5.0, -2.0, 4.0) (0, 0, 0) (0.0, -1.0, 2.0)

Soft Neutral Fill Emerald Highlights

Physical Light ConfigurationsJSON{

"lighting_system": {

"ambient": { "color": "#FFFFFF", "intensity": 0.75 },

"directional": [

{ "id": "key_light", "color": "#FFFFFF", "intensity": 1.2, "position": [5.0, 10.0, 7.0], "cast_shadow": true, "shadow_bias": -0.0001 },

{ "id": "fill_light", "color": "#D1E7DD", "intensity": 0.5, "position": [-5.0, -2.0, 4.0], "cast_shadow": false }

],

"point": [

{ "id": "emerald_accent", "color": "#0F8259", "intensity": 0.8, "position": [0.0, -1.0, 2.0], "distance": 10.0, "decay": 2.0 }

]

}

}

9. Material Librarymat_emerald_glass (Three.js MeshPhysicalMaterial)JavaScriptconst mat_emerald_glass = new THREE.MeshPhysicalMaterial({

color: new THREE.Color("#0F8259"),

transmission: 0.95,

opacity: 1.0,

transparent: true,

roughness: 0.05,

metalness: 0.0,

ior: 1.52,

thickness: 0.8,

attenuationColor: new THREE.Color("#0F8259"),

attenuationDistance: 0.5,

clearcoat: 1.0,

clearcoatRoughness: 0.1,

reflectivity: 0.9

});

mat_ui_glass_panel (CSS Material Rules)CSS.mat-ui-glass-panel {

background: rgba(255, 255, 255, 0.25);

backdrop-filter: blur(20px) saturate(180%);

-webkit-backdrop-filter: blur(20px) saturate(180%);

border: 1px solid rgba(255, 255, 255, 0.40);

box-shadow: 0 20px 40px rgba(0, 0, 0, 0.06), inset 0 1px 0 rgba(255, 255, 255, 0.6);

}

10. Shader LibraryBackground Liquid Shader (shd_background)Vertex ShaderOpenGL Shading Languagevarying vec2 vUv;

void main() {

vUv = uv;

gl_Position = projectionMatrix * modelViewMatrix * vec4(position, 1.0);

}

Fragment ShaderOpenGL Shading Languageuniform float uTime;

uniform vec2 uResolution;

uniform vec2 uMouse;

varying vec2 vUv;

void main() {

vec2 st = gl_FragCoord.xy / uResolution.xy;

vec2 mouseEffect = uMouse * 0.02;

float wave = sin(st.x * 4.0 + uTime * 0.5 + mouseEffect.x) *

cos(st.y * 4.0 + uTime * 0.5 + mouseEffect.y) * 0.05;

vec3 colorStart = vec3(0.98, 0.97, 0.96); // #FAF9F5

vec3 colorEnd = vec3(0.88, 0.87, 0.83); // #E2DFD5

```
vec3 baseColor = mix(colorStart, colorEnd, st.y + wave);
gl_FragColor = vec4(baseColor, 1.0);
}
```

11. Animation System ANIMATION TRIGGER PIPELINE

```
User Action / Scroll ───> GSAP ScrollTrigger
│
┌──────────────────────────────┴──────────────────────────────┐
▼ ▼
DOM Element Transforms WebGL Uniform Updates
(TranslateY, Scale, Opacity) (uTime, Mesh Rotations)
```

Motion RegistryJSON[

```
{
"id": "anim_shatter_implosion",
"target": "obj_3d_x_assembly",
"duration": 1.2,
"ease": "cubic-bezier(0.16, 1, 0.3, 1)",
"from": { "scale": 1.8, "opacity": 0.0, "shards_spread": 4.0 },
"to": { "scale": 1.0, "opacity": 1.0, "shards_spread": 0.0 }
},
{
"id": "anim_card_stagger_enter",
"target": ".service-card",
"duration": 1.0,
"stagger": 0.15,
"ease": "cubic-bezier(0.16, 1, 0.3, 1)",
"from": { "translateY": "80px", "opacity": 0.0 },
"to": { "translateY": "0px", "opacity": 1.0 }
}
]
```

12. Master Timeline===

TIMELINE MARKERS (0.0s - 10.0s)

===

```
[0.0s] ──────────────────────────── [1.2s] ──────────────────────────── [5.5s]
──────────────>
| | |
├─ Shatter Implosion Starts ├─ Hero UI Text Reveals ├─ Services Grid Enters
└─ Camera locked at z=8.0 └─ Camera starts drift └─ Camera Y-Pan (-2.5)

[5.5s] ──────────────────────────── [8.0s] ──────────────────────────── [10.0s]
| | |
├─ Portfolio Horizontal Slide ├─ Process Node Rail Active ├─ Contact Form & Outro
└─ Camera Y-Pan (-5.0) └─ Particles Active └─ Camera Locks (z=7.5)
==========================================================================================
```

13. Scroll ChoreographyComplete Scroll Percentage Mapping ($0\% - 100\%$)JSON[

{
"scroll_pct": "0% - 15%",
"active_section": "Hero Section",
"camera_target_z": 8.0,
"dom_opacity": { "#hero-title": 1.0, "#services-grid": 0.0 },
"webgl_states": { "x_logo_visible": true, "particle_emission": 0.0 }
},
{
"scroll_pct": "16% - 40%",
"active_section": "Services Section",
"camera_target_z": 10.5,
"dom_opacity": { "#hero-title": 0.0, "#services-grid": 1.0 },
"webgl_states": { "x_logo_visible": false, "particle_emission": 0.0 }
},
{
"scroll_pct": "41% - 70%",
"active_section": "Process Node Rail Section",
"camera_target_z": 9.0,
"dom_opacity": { "#services-grid": 0.0, "#process-nodes": 1.0 },
"webgl_states": { "x_logo_visible": false, "tube_rail_visible": true }
},
{

"scroll_pct": "71% - 100%",

"active_section": "Contact Form & Footer Outro",

"camera_target_z": 7.5,

"dom_opacity": { "#process-nodes": 0.0, "#contact-form": 1.0, "#footer": 1.0 },

"webgl_states": { "x_logo_visible": true, "particle_emission": 5000.0 }

}

]

14. Component Librarycomp_navbarDimensions: Width 100%, Height 80px.Layout: Flex Row, justify-content: space-between, align-items: center.Styles: background: rgba(235, 233, 225, 0.75), backdrop-filter: blur(12px).comp_glass_cardDimensions: Width 100%, Padding 32px, Radius 16px.CSS Class: .mat-ui-glass-panel.Hover Behavior:CSS.comp-glass-card {

transition: transform 0.4s var(--ease-quartz), border-color 0.4s ease;

}

.comp-glass-card:hover {

transform: translateY(-8px) scale(1.01);

border-color: rgba(255, 255, 255, 0.80);

}

15. Interaction State Machines GLASS CARD INTERACTION STATE MACHINE

```
[IDLE STATE] ─── mouseenter ───> [HOVER STATE] ─── mousedown ───> [PRESSED STATE]
▲ | |
| | mouseleave | mouseup
└──────────────────────────┴──────────────────────────────┘
```

JSON{

"state_machine_card": {

"idle": { "translateY": "0px", "scale": 1.0, "border_opacity": 0.4 },

"hover": { "translateY": "-8px", "scale": 1.01, "border_opacity": 0.8 },

"pressed": { "translateY": "-2px", "scale": 0.99, "border_opacity": 1.0 }

}

}

16. Cursor SystemConfiguration BlueprintOuter Ring: Radius 20px, Border 1.5px rgba(24,25,28,0.4), Lag Factor $\text{lerp} = 0.12$.Inner Dot: Radius 3px, Color #0F8259, Lag Factor $\text{lerp} = 0.02$.JavaScript// Cursor Position Lerp Loop

function renderCursor() {

targetX += (mouseX - targetX) * 0.12;

targetY += (mouseY - targetY) * 0.12;

cursorRing.style.transform = `translate3d(${targetX}px, ${targetY}px, 0)`;

requestAnimationFrame(renderCursor);

}

17. Particle SystemGPUParticleSystem SpecificationCapacity: $5,000$ active particles.Color Range: Blend between #0F8259, #10B981, and #A7F3D0.Particle Size: $2.0\text{px} - 6.0\text{px}$ randomized per particle.Velocity Vector: Spiral curve around Y-axis:$$\vec{v} = \begin{pmatrix} -\sin(t) \cdot r \\ 0.5 \\ \cos(t) \cdot r \end{pmatrix}$$18. Responsive BlueprintBreakpoint MatrixTokenBreakpoint RangeGrid ColumnsTypography Scale FactorCamera Distance ZDesktop$> 1280\text{px}$12 Columns$1.0\times$$8.0$Laptop$1024\text{px} - 1279\text{px}$12 Columns$0.9\times$$8.5$Tablet$768\text{px} - 1023\text{px}$6 Columns$0.8\times$$9.5$Mobile$< 767\text{px}$4 Columns$0.68\times$$11.0$19. AccessibilityReduced Motion: Standard fallback for users with prefers-reduced-motion: reduce. 3D animations are bypassed; camera movements are disabled; DOM elements fade in static locations.Color Contrast: All body text meets WCAG AAA standards ($> 7:1$ contrast ratio against light backdrop).Keyboard Navigation: All interactive cards and buttons expose visible focus rings (2px solid #0F8259).20. Performance Budget GPU FRAME TIME BUDGET (16.6ms Target)

```
┌──────────────────────────────┬──────────────────────────────┬──────────────────────────────┐
│ WebGL Render (6.0ms) │ GLSL Post-Process(4.5ms)│ DOM Composite (3.0ms) │
└──────────────────────────────┴──────────────────────────────┴──────────────────────────────┘
```

[====================================== 13.5ms ======================================>]

[Margin: 3.1ms]

Quantitative ConstraintsTarget FPS: 60 FPS constant.Maximum Triangle Count: $< 50,000$ active triangles per frame.Maximum Draw Calls: < 35 draw calls per frame.Texture Memory Limit: $< 64\text{MB}$.21. Implementation BlueprintFile Architecture Plansrc/

```
├── app/
│ ├── layout.tsx
│ └── page.tsx
├── components/
│ ├── dom/
│ │ ├── Navbar.tsx
│ │ ├── Hero.tsx
│ │ ├── ServicesGrid.tsx
```

```
│ │ └── ContactForm.tsx
│ └── canvas/
│ ├── SceneCanvas.tsx
│ ├── GlassXMesh.tsx
│ ├── ProcessRail.tsx
│ └── ParticleSystem.tsx
├── shaders/
│ ├── background.fragment.glsl
│ └── glass.fragment.glsl
└── styles/
└── globals.css
```

22. QA Checklist[ ] Visual Accuracy: Background color matches #EBE9E1 across DOM and WebGL canvas elements.[ ] Refraction Pass Verification: Emerald glass ($IOR = 1.52$) renders visible refraction offset against background text.[ ] Frame Rate Validation: Test maintaining stable 60 FPS performance under 4K resolution displays.[ ] Scroll Synchronization: Verify GSAP ScrollTrigger timeline pins correctly without jitter across all breakpoints.Single Source of Truth SignatureDocument Version: 2.2.0-MASTER-FINALStatus: Approved for ImplementationTarget Stack: Next.js + React Three Fiber + GSAP + TailwindCSS